
Problem Set 1 Solutions

1. (**23 points**) The pro-school choice organization EdChoice conducts a regular opinion survey on matters related to school enrollment. You can find highlights and detailed results from the July 2021 survey on Github: <https://github.com/spcorcor18/LP0-8800/tree/main/assignments/PS01>. For your information, the most recent results are found here: <https://edchoice.mcdatahub.com/edchoice/>. Use the 2021 reports to answer the questions below.

- (a) The EdChoice report contains many interesting statistics, including items related to support for school choice, teacher salaries, vaccination, and in-person schooling during COVID-19. Based on a quick skim of this report, would you say these are intended as descriptive or inferential statistics? Briefly explain your answer. (**3 points**)

These are intended to be inferential statistics. This is clear since the data were collected via a “national sample of adults” and a margin of error (“measure of precision”) was reported. The margin of error is an example of “quantifying the uncertainty” associated with an inferential statistic, as described in class.

- (b) Consider the data the authors used for this report. What was the unit of observation? Was the data a population or a sample? How were the data obtained? How many observations were in the data? (**4 points**)

The unit of observation was an adult respondent to a survey conducted online. The data were a *sample* of 2,200 adults and 1,228 school parents.

- (c) Consider the following list of measures used in this report. For each, state whether the measure is: (1) quantitative or categorical; (2) nominal, ordinal, or interval/ratio scale. Base your answer on how the data are originally collected by the survey and provide a brief explanation if necessary. (**6 points**)

- i. Feelings about whether K-12 education is generally going in the right direction: **categorical, nominal**. Respondents were given the choice between “generally going in the right direction” or “generally gotten off on the wrong track.” One could also argue this is an ordinal scale since the aim is to

measure the extent to which individuals think education is going in the right direction.

- ii. How much trust one has in principals to make good decisions about education: **quantitative, ordinal**. Choices were ordinal, including “a lot,” “some”, “not that much”, and “not at all”.
- iii. If given the option, what type of school parents would select for their child: **categorical, nominal**. Options appear to have included private school, regular school, home school, charter school, and “don’t know”.
- iv. School parents’ comfort with their children returning to school in person: **quantitative, ordinal**. Choices were ordinal, including “very comfortable,” “somewhat comfortable”, “not that comfortable”, and “not at all comfortable”. (An “I don’t know” option was also provided).
- v. Parents’ opinion on homeschooling since COVID-19 began: **quantitative, ordinal**. Similar Likert scale to the trust and comfort items above.
- vi. How much parents would be willing to spend on a monthly basis to participate in a “learning pod”: **quantitative, ratio**. Respondents provided a dollar amount.

(d) What is an example of a dichotomous variable used in this report? (2 points)

One example is the question about general feelings about K-12 education. There were two possible answers: “generally going in the right direction” or “generally gotten off on the wrong track.”

(e) Take a look at Table EC1_1 in the detailed report. Which subgroups had the most positive view of K-12 education? Which subgroups had the most negative view? Do you see any interesting differences in the responses by gender, race, income, or political party? (4 points)

Groups that had the most positive view of education nationally included Democrats (61%), Liberals (63%), Labor Union members (64%), Blacks (60%), private school parents (65%), and special education parents (57%). Those with the most negative view included those age 65+ (30%), Republicans (33%), conservatives (33%), high school parents (38%), small town/rural area residents (34-35%).

- (f) What is the population of interest in this study, the sampling frame, and sampling method? Briefly summarize. Note: more details on the report's methodology are provided here: <https://edchoice.morningconsultintelligence.com/methodology/> (4 points)

There are several populations of interest in this study, including all U.S. adults, parents of school-aged children, and teachers. An important thing to note about the survey is that it uses *non-probability* sampling. This term means that the sample was not drawn from a well-defined frame from which one might calculate the probability of obtaining various samples. The survey is internet-based and respondents self-select into participating. (Interestingly, Morning Consult—the survey company—bids for a stratified sample of respondents from a large exchange of survey-takers). Weights are used to make the resulting sample “look” like the population of interest.

2. (6 points) A sociologist wants to estimate the average age at marriage for women in New England in the early 18th century. She finds within her state archives of marriage records for a large Puritan village for the years 1700-1730. She then takes a sample of those records, noting the age of the bride for each. The average age in these records is 24.1 years. Using a statistical method covered later, the sociologist estimates the average age of bridges at marriage for the population to be between 23.5 and 24.7 years.

- (a) What part of this example is *descriptive*? The descriptive part of this example is the average age of marriage in the sample (24.1).
- (b) What part of this example is *inferential*? The inferential part of this example is the estimate of the average age of brides in the population (23.5 to 24.7).
- (c) To what population does the inference refer? The population of interest is women who married in New England in the early 18th century.

3. (3 points) Classify the following variables as either categorical or quantitative. For the quantitative variables, explain whether and why the measurement scale is (likely) discrete or continuous.

- (a) Number of students enrolled at Vanderbilt University in any particular term: quantitative and discrete. While the number of students is definitely a discrete measure, in practice we would analyze it as a continuous variable, since it takes on many values.
- (b) Distance an individual can run in five minutes: quantitative and continuous.
- (c) Hair color: categorical.
- (d) State (from 1=Alabama to 50=West Virginia): categorical. The numerical values assigned here have no inherent meaning.
- (e) Self-concept as measured by the degree of agreement with the statement, "I feel good about myself," on a five point (Likert) scale: categorical (and ordinal). While the underlying construct of "self-concept" can be thought of as continuous (and in theory, measured), it is operationalized here as an ordinal categorical variable.
- (f) Lack of coordination as measured by the length of time it takes an individual to assemble a puzzle: quantitative and continuous.